

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials
Total training paths	16 · 524 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Accel
Sensor columns	78 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4212
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=2095 · test=105

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	1.0000	0.0000	0.8952
LDA	1.0000	0.0000	0.8095
SVM (Linear)	1.0000	0.0000	0.7810
SVM (RBF)	1.0000	0.0000	0.7429
Logistic Regression	1.0000	0.0000	0.7429
KNN	0.9823	0.0125	0.5810

Per-class hold-out F1

Class	Random Forest	LDA	SVM (Linear)	SVM (RBF)	Logistic Regression	KNN	Support
Double_Closed_Fist	0.909	0.788	0.667	0.645	0.667	0.562	16
Double_Nothing	0.609	0.571	0.667	0.519	0.500	0.400	14
Double_Okay	1.000	0.815	0.828	0.800	0.741	0.474	15
Double_Open_Palm	1.000	0.778	0.848	0.839	0.800	0.688	15
Double_Phone	0.857	0.774	0.667	0.643	0.667	0.400	15
Double_Pistol	0.909	0.903	0.867	0.897	0.897	0.839	15
Double_Spiderman	0.909	0.968	0.903	0.824	0.857	0.667	15
macro avg	0.885	0.800	0.778	0.738	0.733	0.576	105

External test (NewTestData)

Algorithm	External test acc
Logistic Regression	0.7429
SVM (Linear)	0.6286
Random Forest	0.6286
SVM (RBF)	0.6000
LDA	0.5429
KNN	0.3143

Per-class external accuracy

Class	Logistic Regression	SVM (Linear)	Random Forest	SVM (RBF)	LDA	KNN	Support
Double_Closed_Fist	0.400	0.400	0.400	0.600	1.000	0.600	5
Double_Nothing	0.800	0.600	0.600	0.800	0.600	0.000	5
Double_Okay	0.600	0.600	0.400	0.400	0.400	0.400	5
Double_Open_Palm	1.000	1.000	1.000	0.800	0.800	0.200	5
Double_Phone	0.400	0.000	0.200	0.000	0.000	0.000	5
Double_Pistol	1.000	0.800	0.800	0.600	0.200	0.400	5
Double_Spiderman	1.000	1.000	1.000	1.000	0.800	0.600	5